

EP06 — How does a computer store a photo?

HINGLISH (Roman)

01 Title

Computer ek photo **store** kaise karta hai? Welcome to Katixo KhojLab. Aap photo kheechte ho, wo gallery mein aa jaati hai — par machine ke liye wo photo bas numbers ka ek **grid** hai. Aaj zoom in karke unhe dekhte hain.

[CUE] Title pe ~2s ruko, calm curious tone, phir aage badho.

02 The Hook

Koi bhi photo dekho. Wo smooth lagti hai, real, colour se bhari. Par twist ye hai — computer ko koi "picture" dikhti hi nahi. Use dikhta hai numbers ka ek **bahut bada spreadsheet**. Toh chalo zoom in karte hain, aur aur paas, jab tak aap un numbers ko khud na dekh lo.

[CUE] Yahan energy high — ye hook hai. "zoom in" pe pinch-zoom ka mime.

03 Aaj ka sawaal

Toh aaj ka sawaal. Aapki selfie colour aur light hai. Par computer sirf **numbers** samajhta hai. Toh aakhir wo poori photo apne andar rakhta kaise hai? Chalo pata karte hain.

[CUE] Thoda slow. Sawaal ko land hone do, phir aage.

04 Idea 1 — Pixels

Idea number one. Har image chhote-chhote coloured squares ka ek grid hai, aur har square ko **pixel** kehte hain. Kisi bhi photo mein kaafi zoom in karo, aur aapko wo squares dikhne lagte hain. Ek phone photo mein inke **millions** hote hain. Yahi to **"megapixels"** ka matlab hai — "mega" bas ek fancy shabd hai **million** ke liye. Baarah-megapixel camera ek shot mein baarah million pixels capture karta hai.

[CUE] pixel chips ek-ek karke reveal karo, phir "megapixels" pe land karo.

05 Idea 2 — RGB

Idea two. Har pixel bas ek colour hai — aur har colour **teen numbers** mein store hota hai: Red, Green, aur Blue. Har ek **zero se 255** tak jaata hai, yaani poora off se poora bright. Two-fifty-five, zero, zero — pure red. Zero, zero, zero — saari lights off — black. Aur teeno 255 pe — white. In teen numbers ko mix karo aur screen pe **koi bhi colour** bana lo.

[CUE] RGB triples ko code block se padho; 0 aur 255 pe zor.

06 Idea 3 — Numbers ki badi list

Ab sab jodo. Ek photo millions of pixels hai, aur har pixel teen numbers — toh poori photo numbers ki ek **bahut badi list** ban jaati hai. Aur pichhle episode se yaad rakho, har number aakhir mein **binary** mein store hota hai — wahi zeros aur ones. Isiliye photos **"heavy"** hoti hain. Zyada pixels matlab zyada numbers matlab bada file — isiliye photos ko hum **megabytes**, yaani MB, mein naapte hain.

[CUE] "binary" pe 0s-and-1s wale episode ka callback nod.

07 Idea 4 — Compression

Idea four. Har ek number store karna waste hai — toh hum **compression** se hoshiyaari karte hain. **JPEG** jaisa format do smart cheezein karta hai: wo aise chhote details phenk deta hai jo aapki aankh waise bhi miss nahi karti, aur repeated patterns ko baar-baar likhne ki jagah unhe reuse karta hai. Isiliye ek JPEG raw image se **kaafi chhota** ho jaata hai. Trade-off ye hai: **PNG** sab kuch rakhta hai — wo lossless hai. JPEG thoda detail drop karta hai space bachaane ke liye — wo lossy hai.

[CUE] "lossless / lossy" pe PNG-vs-JPEG note card reveal karo.

08 Why it matters

Aur achaanak aapki poori gallery samajh aane lagti hai. **Megapixels** batate hain camera kitne pixels pakadta hai. **File size** batata hai kitne numbers store karne pade. Koi purani, chhoti photo kholo ya bahut zyada zoom in karo, toh squares dikhne lagte hain — wo "**blocky**" ya pixelated look bas pixels ka bada ho jaana hai. Aur jab logo share karo, PNG chuno taaki crisp rahe; normal photo ke liye JPEG space bachaata hai.

[CUE] megapixels / size / blocky / PNG bolte waqt har bullet ko point karo.

09 Common myth

Aur isse aata hai myth. Log sochte hain computer actual colours store karta hai — ki asli picture kisi tarah file ke andar baithi hai. Aisa nahi hai. Wo **numbers** store karta hai — har pixel ke red, green, blue values. Aapki **screen** hi un numbers ko wapas light mein badalti hai. "Photo" sach mein sirf tab exist karti hai jab wo **display** hoti hai.

[CUE] Myth pe ek beat, phir green "Reality" card reveal karo.

10 Recap

Quick recap. Ek: ek photo **pixels** ka grid hai, aur har pixel bas kuch numbers — uska red, green, aur blue. Do: **compression** file ko chhota karta hai, hoshiyaari se wo detail drop karke jo aap notice bhi nahi karoge.

[CUE] Crisp aur clear — ye do lines log yaad rakhein.

11 CTA

Bas itna hi — aapki photo sach mein numbers ka ek bada grid hai, aur ab aap unhe dekh sakte ho. Agar isse aapki photo dekhne ki nazar badal gayi, toh Katixo KhojLab ko subscribe karo. Agla sawaal: "the cloud" actually hota kya hai? Milte hain agle video mein.

[CUE] Warm, friendly. Subscribe line pe awaaz mein smile.
